



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)



Course: Mathematics I
Course Code: 0705210101

Symbiosis Institute of Technology, Nagpur

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Assistant Professor (Sr. Scale)



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University Mission

- Promoting International understanding through quality education.

University Vision

- To inculcate spirit of 'Vasudhaiva Kutumbakam' (the world is one family).
- To contribute towards knowledge generation and dissemination.
- To promote ethical and value-based learning.
- To foster the spirit of national development.
- To inculcate cross cultural sensitization.
- To develop global competencies amongst students.
- To nurture creativity and encourage entrepreneurship.
- To enhance employability and contribute to human resource development.
- To promote health and wellness amongst students, staff & community.
- To instill sensitivity amongst the youth towards the community and environment.
- To produce thought provoking leaders for the society.



Course Outcomes and Program Outcomes

- Apply successive differentiation to find nth derivative of product of two functions, to evaluate limits of indeterminate forms and to find the series expansion of functions.
- Apply the concepts of partial differentiation to solve problems on homogeneous functions, Jacobians and maxima & minima.
- Find the rank of a matrix and solve system of simultaneous equations.
- Find Eigen values and Eigen vectors, apply Caley-Hamilton theorem to find inverse and higher powers of a matrix and transform matrix to diagonal form.



Evaluation Methods/Plan

Sr. No.	Component	Syllabus	Max. Marks (30)	Tentative Date		
				Notification	Conduction	Result Declaration
1	CA1: Open Book Test	Unit I	10	3 rd week of Aug	Aug 25-30	Within 15 Days
2	CA2: Mid-Test (Centralized)	Unit II	10	2 nd week of Sept	Sept 27-30	Within 15 Days
3	CA3: Assignment Test	Unit III	10	3 rd week of Oct	Oct 29-31	Within 15 Days

Eligibility to appear for CA Component: Must have **attendance above 75%** before the scheduled CA components.



Attendance criteria

- **Eligibility to appear for CA Component:** Must have **attendance above 75%** before the scheduled CA components.
- To be eligible for appearing in the end-semester exam, students must maintain a minimum attendance of 75%.
- Students failing to meet the attendance criteria will receive a "Course Not Granted" (CNG) status.
- CNG status indicates ineligibility to take the end-semester exam.



Syllabus of the subject [Credits: 3, Max. Marks: 75 (CA: 30 + ESE: 45), Contact Hours: 2+1]

- ❑ **Differential Calculus:** Successive differentiation, Leibnitz's theorem, Taylor's and Maclaurin's expansions, Indeterminate forms, L'Hospital's rule
- ❑ **Partial Differential Calculus and applications:** Partial differentiation, Euler's Theorem for homogeneous functions, Total derivatives, Jacobians and its properties, Maxima and minima of functions of two variables, Lagrange's method of multipliers
- ❑ **Matrices, vector spaces and sub-spaces:** Rank of a matrix: Echelon form, Solution of system of equations, Eigenvalues, Eigenvectors, Properties of eigenvalues and eigen vectors, Cayley-Hamilton, theorem and its applications, Euclidian spaces, Vector spaces, Subspaces, Linear Combination, Linearly dependent and independent vectors, Basis, Dimension, Change of Basis, Linear transformations, Matrix representation of linear maps. Rank and nullity of a linear transformation, Orthogonality





Academic Calendar

Symbiosis Institute of Technology, Nagpur Campus														
Academic Calendar ODD Semester 1st Sem														
Session: 2025-26														
	Mon	Tue	Wed	Thur	Fri	Sat	Sun	Days	Holidays	List of Activities	Working days	C A		
												Credit 2	Credit 3	Credit 4
Jul-25		1	2	3	4	5	6			July 15-29 Induction program, 30 July Commencement and Registration	2			
	7	8	9	10	11	12	13							
	14	15	16	17	18	19	20							
	21	22	23	24	25	26	27							
	28	29	30	31				2						
Aug-25					1	2	3	2	August 9 Rakshabandhan, 15 Independence and 27 Ganesh Chaturthi		23		CA1 Aug 25-Aug 30	CA1 Aug 25-Aug 30
	4	5	6	7	8	9	10	5						
	11	12	13	14	15	16	17	5						
	18	19	20	21	22	23	24	6						
	25	26	27	28	29	30	31	5						
Sep-25	1	2	3	4	5	6	7	4	Sept 5 and 6 : Eid-e-Milad and Anant Chaturdashi	Sep 25-26 : Student Mid-Semester Feedback	21	CA1/ MSE Sep 27-30	CA2/ MSE Sep 27-30	CA2/ MSE Sep 27-30
	8	9	10	11	12	13	14	6						
	15	16	17	18	19	20	21	6						
	22	23	24	25	26	27	28	5						
	29	30												
Oct-25			1	2	3	4	5	3	Oct 2 Gandhi Jayanti and Oct 17-23: Diwali	Oct 10-11: Sports Day, Oct 1: Open House	20	CA2 Oct 29-31	CA3 Oct 29-31	CA3 Oct 29-31
	6	7	8	9	10	11	12	6						
	13	14	15	16	17	18	19	4						
	20	21	22	23	24	25	26	2						
	27	28	29	30	31			5						
Nov-25						1	2	1		Nov 26- Dec 2: Practical End Sem Exam & Student End Semester Feedback,	25			CA4 Nov 13-15
	3	4	5	6	7	8	9	6						
	10	11	12	13	14	15	16	6						
	17	18	19	20	21	22	23	6						
	24	25	26	27	28	29	30	6						



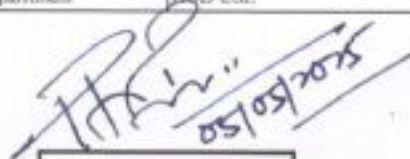
Academic Calendar

Dec-25	1	2	3	4	5	6	7	2	Dec 25: Christmas	ESE: Dec5-13	2			
	8	9	10	11	12	13	14							
	15	16	17	18	19	20	21							
	22	23	24	25	26	27	28							
	29	30	31											
									Total Days:	93				

Particulars	Start Date	End Date	
Teaching	30-07-2025	02-12-2025	93
PL	03-12-2025	04-12-2025	2
ESE	05-12-2025	13-12-2025	9
Semester Break	14-12-2025	31/12-2025	18
Commencement of Next Sem (2nd Sem)	02-01-2026		

(Verified)	(Verified)
Signature: <i>[Signature]</i>	Signature: <i>[Signature]</i>
Name: Dr. P. V. M. (HAWAW)	Name: Dr. S. S. Badhye
Date: 5-5-2025	Date: 05-05-2025
Examination Department	HOD CSE

	Commencement of Semester
	SUNDAYS
	HOLIDAY as per SIU
	CA(Credit 4)
	CA(Credit 3)
	CA(Credit 2)
	MSE/ESE
	ESE Practical
	PL
	Last Teaching Day
	Open House
	Cultural/Technical Fest



Director's Signature
Director
Symbiosis Institute of Technology
Nagpur





Scope of the subject in Computer Science & Engineering, its prerequisite and other related subjects.

Scope of Mathematics-I in Computer Science & Engineering (CSE)

Mathematics-I provides foundational tools and concepts that are essential for many core and advanced areas in Computer Science & Engineering. The topics covered form the basis for algorithm design, data analysis, graphics, machine learning, and more.

- ❑ **Differential Calculus Scope in CSE:** Essential for algorithmic analysis, especially for performance optimization.

Used in computer graphics for curve modeling, motion, and transformations, Crucial in machine learning (e.g., gradient descent).

Applications: Rate of change analysis, optimization problems, real-time system responsiveness.

- ❑ **Partial Differential Calculus and Applications Scope in CSE:** Foundational in image processing, computer vision, and scientific computing, Applied in simulation (e.g., heat, wave, and fluid simulations in game development and animation), Critical in AI/ML for optimization and modeling of multivariable systems.

Applications: Feature extraction in images, probabilistic models, neural network training.

- ❑ **Matrices, Vector Spaces, and Linear Algebra Scope in CSE:** Linear algebra is the backbone of machine learning, deep learning, and data science, Matrices are used in graph theory, computer graphics, cryptography, and data compression, Eigenvalues and eigenvectors are fundamental in PCA (Principal Component Analysis), Google's PageRank, and recommendation systems.

Applications: Solving systems of equations in simulations, dimensionality reduction in big data, matrix transformations in graphics.



Scope of the subject in Computer Science & Engineering, its prerequisite and other related subjects.

Prerequisites for Mathematics-I

- ☐ To understand and apply the topics in this syllabus, students should be comfortable with:
- ☐ Basic **differentiation and integration** (from high school calculus)
- ☐ Algebraic manipulation and functions
- ☐ Basic understanding of geometry
- ☐ Logical thinking and analytical reasoning



Scope of the subject in Computer Science & Engineering, its prerequisite and other related subjects.

Other Related Subjects

- ☐ **Discrete Mathematics:** Logical reasoning, set theory, combinatorics – foundational for algorithms and graph theory.
- ☐ **Data Structures & Algorithms:** Time complexity analysis, recursion, and optimization use calculus and logical formulation.
- ☐ **Computer Graphics:** Uses matrix algebra, vector spaces, eigenvalues for transformations, projections, and rendering.
- ☐ **Machine Learning:** Linear algebra (matrices, eigenvectors), calculus (gradient descent, optimization), multivariable functions.
- ☐ **Artificial Intelligence:** Optimization techniques from calculus, logic, and linear algebra models.
- ☐ **Numerical Methods / Numerical Analysis:** Solving differential equations, interpolation, matrix solutions to systems.
- ☐ **Signals and Systems:** Uses differential equations, linear algebra, and transforms.
- ☐ **Digital Image Processing** Partial derivatives for edge detection, Jacobians for transformations.
- ☐ **Robotics and Control Systems:** Partial differential equations and matrix operations in motion planning and feedback systems.
- ☐ **Cryptography and Network Security:** Linear algebra for encoding/decoding, modular arithmetic, matrix-based ciphers.
- ☐ **Big Data Analytics / Data Science:** Uses matrices, vector spaces, and optimization for modeling and prediction.
- ☐ **Compiler Design:** Optimization problems during code generation and memory allocation.
- ☐ **Computer Networks:** Uses queueing theory and optimization techniques derived from calculus.
- ☐ **Operations Research:** Optimization techniques, Lagrange multipliers, linear programming.



Time Table

Symbiosis Institute of Technology, Nagpur										
Ver. 1.1			Time Table B. Tech. (Computer Science and Engineering)					Session: 2025-2026		
								w.e.f.: I Sem: 30/07/2025		
PERIOD → DAYS ↓	Sem	9.05 - 9.55 (1)	9.55 - 10.45 (2)	10:50 - 11:40(3)	11:40- 12:30 (4)	12:30 - 01:20 (5)	01:20 - 02:10 (6)	02:10- 03:00 (7)	03:00 - 3:50 (8)	03:55 - 04:45 (9)
MON	I									
TUE	I									
TUE	I									
WED	I									
WED	I									
THU	I									
FRI	I									
FRI	I									
Sat										



About yourself and your area of research

Dr. Praveen Kumar Dhankar

Education: Ph. D. (Mathematics), CSIR-NET (11th AIR)

Experience: 11 + Years

Professional Affiliations:

Professional Bodies: FRAS, IOP, IAU, LMS, IUCAA Visiting Researcher, BOM-KKSU, Ramtek

Expert Area: Relativistic Cosmology and Mathematical Physics, Applied Mathematics.

Research Focus: Modified Theories of Gravity, Bayesian Statistics, Data Analysis, Machine Learning

Publications: Journals: 15, Conferences: 00, Books: 01

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Thank
You!

